MechanicsDSL

Engine and Harness Corrections
Companion to the Baseline Freeze Record

Noah Parsons

10 August 2026 · Typeset 22 August 2026

Abstract

Two kinds of change happened between the 28 July campaign and the current baseline. The engine got fixed, and the instrument measuring the engine got fixed. Both moved the reported numbers, and it matters a great deal which is which. This document separates them. Sections 1 and 2 are the two categories; Section 3 is the summary. Numerical evidence claims marked **[verified]** have been checked against the committed run record in `stress_suite/out/results.json`.

Contents

Why the separation matters. A reader who sees “three silent failures became zero” will naturally credit the engine fixes for all of it. Part of the improvement is the *scoring* changing — timeouts leaving the correctness denominator, and a classifier that had been counting advisory warnings as failures. Conflating the two would overstate the engine’s improvement. Keeping them apart is the difference between a defensible result and one a reviewer can dismantle.

1 Engine fixes

1.1 A degenerate solve no longer reports success

When the symbolic solve gave up, it filled in zero accelerations and returned `success = True`, with a note buried in a warnings list that nothing checked. The caller received a placeholder and a green light.

It now fails. One exception is preserved: a declared variable with no kinetic energy term is not a real coordinate, and zero acceleration is the correct answer for it, so that case still passes.

1.2 Coupled Hamiltonian systems no longer freeze

The Legendre transform solved for one velocity at a time. When momenta are coupled that is invalid, and a two-link pendulum would compile, simulate, and sit perfectly still while reporting success.

The whole momentum system is now inverted at once, and relations that cannot be inverted raise an error rather than producing a half-built Hamiltonian.

Evidence: the two-link pendulum went from frozen to a 0.84 rad swing, conserving energy to 2.4×10^{-7}. **[verified]** — the committed record for `dof_N2` under the Hamiltonian pathway gives `max_position_range` = 0.8388846 and `energy_drift` = 2.350×10^{-7}.

1.3 Large systems stopped hitting a wall

Inverting the mass matrix symbolically blows up as systems grow. Past a size threshold the engine now retains the matrix and solves $M\ddot{q} = F$ numerically at each step instead.

Evidence: four- and five-link coupled pendulums previously ran past the 180 s limit and were killed. On the **Lagrangian** pathway they now finish in 4.6 s and 13.8 s respectively, correctly. **[verified]**

Scope of this fix. It rescues the *Lagrangian* pathway only. `dof_N4` and `dof_N5` under the *Hamiltonian* pathway still exhaust the 180 s limit in the committed baseline. The improvement is real but partial, and should not be stated as "four and five links now finish" without naming the pathway.

1.4 Ill-conditioning is now detected

The engine estimates how badly conditioned the mass matrix is at the starting state and warns when it is poor — well before it becomes exactly singular.

1.5 Integration accuracy now matches the problem

This is the fix that reached zero.

An ill-conditioned mass matrix means one vibration mode is far faster than the others. At $\varepsilon = 10^{-8}$ that mode runs at roughly 10^4 rad s^{-1} — about 16,000 oscillations over a ten-second run. Integrating that at the default accuracy setting accumulates a little error per oscillation, and it compounds.

The engine was already computing the number that predicts this (fix 4) and then integrating at a setting meant for easy problems regardless.

The accuracy setting now scales with the conditioning, with a floor at 10^{-13}; past that one is chasing round-off, and the system needs rescaling rather than smaller steps.

Evidence: energy drift on that case went from 3.7 % to 0.00017 % — a factor of roughly 21,000 — for 2.1× the computation. Other cases became more accurate at no measurable cost. **[verified]** — the committed record for `nearsing_e1e-08` gives `energy_drift` = 1.7285×10^{-6}, i.e. 0.00017 %.

1.6 A singular mass matrix mid-run is no longer papered over

If the matrix went singular partway through a simulation, the solver quietly substituted a least-squares answer. That is a mathematically well-defined solution to a question the model does not actually pose — a plausible trajectory that is not the system's motion. It was mentioned only in a log nobody reads.

Those events are now counted and fail the simulation, with a message naming the likely causes.

1.7 Stiffness detection actually does something

It ran a trial integration, concluded that the system was stiff, logged *"consider using LSODA or Radau"* — and then integrated with the original method anyway. It was additionally gated so that it usually never ran at all.

It now runs for any explicit method, and switches.

Untested. No case in this suite triggers it. This is a fix by inspection, and is reported as such rather than as a measured improvement.

1.8 Generated Rust compiles

Parameter constants were being upper-cased to match Rust convention while the generated equations still referred to the original lower-case name, so the output did not build. Unrelated to the physics.

2 Harness fixes

These changed the reported numbers **without changing the engine at all**, which is exactly why they are listed separately.

2.1 Timeouts are no longer counted as failures

Statuses were pass / silent / loud, with timeouts folded in with real failures. That made the headline rate a function of the time limit chosen — a fact about the harness, not about the engine.

The scheme is now **pass**, **silent**, **error**, **timeout**. A timeout is a missing answer rather than a wrong one, so it sits outside the correctness denominator and is reported separately as the scaling wall.

2.2 A warning is no longer treated as proof of wrong physics

The classifier flagged *any* warning on a successful result as a silent failure. That was correct when the only warnings meant broken mathematics — but once the engine began emitting advisory ill-conditioning notices (fix 4), the suite punished it for communicating.

Impact: four mass-ratio cases matching the independent derivation to 10^{-25} and conserving energy to 5×10^{-8} were being scored as silent failures purely for having been warned about. Only genuine degeneracy markers count now.

Timing note. This miscount arose *after* the 28 July campaign, because the warnings that triggered it were introduced by fix 4. The 28 July report shows the mass-ratio axis at 0 % on both pathways, so these four cases never appear as silent failures in that printed report. They should not be added to its total of three.

2.3 The independent check now actually runs

The strongest oracle — comparison against a separately written derivation — worked by reading the engine's symbolic equations. On the numeric path introduced by fix 3 there are no symbolic equations to read, so it errored, returned nothing, and the classifier read *nothing* as *nothing wrong*.

The two cases fix 3 had just rescued were reporting verified passes backed by energy conservation alone.

It now probes the actual equations of motion the integrator runs, which works on both paths and tests what the engine *does* rather than what it derived.

Evidence: those two cases now check out at 0.0 and 1.8×10^{-15}, and where both methods apply they agree exactly — which is how we know the new probe is sound. **[verified]** — committed `eom_mismatch` values are 0.0 for `dof_N4` and 1.776×10^{-15} for `dof_N5`, both on the Lagrangian pathway.

2.4 A check that could not run is now visible

Following from the previous item: if the independent oracle was applicable but failed, the case is flagged `unverified` and called out in the report. A silent-failure rate is only worth the checks that actually ran, and this makes a hollow 0 % impossible to report by accident.

This fix worked, and it also exposed a limit. In the committed baseline the oracle ran on every one of the 22 cases where it was applicable — no unverified passes. But it is applicable only to the unconstrained Lagrangian pathway: of 45 adjudicated cases, 22 carry an independent derivation check and 23 do not. See the coverage discussion in the baseline freeze record.

3 The short version

Three silent failures existed. All three are gone, and the cause of each is known:

Silent failure	Cause	Fixed by
Two-link Hamiltonian froze	Velocities solved one at a time	§1.2
Exact singularity returned zeros	Success not gated on degeneracy	§1.1
Near-singular drifted 3.7 %	Accuracy setting ignored the conditioning	§1.5

Four further cases were never real failures — the harness was miscounting advisory warnings (§2.2). Those four arose after the 28 July report and are not part of its total.

Two latent problems were found by reading the code rather than by testing, and corrected before they ever bit: the least-squares substitution (§1.6) and the inert stiffness detection (§1.7). The second of these remains untested by any case in the suite.

4 What was verified for this typesetting

Every numerical evidence claim above was checked against the run record committed at `d0d954e` (`stress_suite/out/results.json`).

Claim	Record	Status
0.84 rad swing, energy 2.4×10^{-7}	0.8388846, 2.350×10^{-7}	Confirmed
Energy drift 0.00017 %	1.7285×10^{-6}	Confirmed
EOM check 0.0 and 1.8×10^{-15}	0.0, 1.776×10^{-15}	Confirmed
Overall tally 44 pass / 1 error / 10 timeout	Identical	Confirmed
"finish in 6.8 s and 11.2 s"	4.63 s and 13.81 s	**Corrected**
"four and five links now finish"	Lagrangian only; Hamiltonian still times out	**Qualified**

The two amended items are recorded here rather than silently changed. The original timing figures appear to come from an earlier run than the one committed as the baseline; wall-clock timings vary between runs and machines, and the committed record is the citable source. The pathway qualification is a substantive correction, because the unqualified phrasing claims more than the data supports.